

1. Executive Summary

This Business Requirement Document (BRD) defines the objectives, scope, data requirements, KPIs, and deliverables for the HR Analytics & Employee Attrition Dashboard project.

A mid-sized organization is experiencing an employee attrition rate of approximately 16%, leading to increased recruitment costs, knowledge loss, and reduced team productivity. The HR leadership team requires a data-driven dashboard to identify attrition patterns, understand root causes, and take proactive retention measures.

This project covers the full analytics lifecycle — from data cleaning and SQL analysis to Power BI dashboard development — using the IBM HR Analytics dataset as the simulation of real organizational data.

2. Business Problem Statement

The organization currently lacks a centralized view of workforce health metrics. HR decisions are made based on manual reports and intuition rather than data. Key pain points include:

- No visibility into which departments, roles, or age groups have the highest attrition
- No understanding of how salary, overtime, job satisfaction, and work-life balance correlate with employee turnover
- Manual headcount tracking in spreadsheets leads to delayed and inconsistent reporting
- HR team cannot predict which employees are at risk of leaving before they resign

Target outcome:

Reduce overall attrition rate from ~16% to below 12% within 2 quarters by providing HR managers with actionable insights through an interactive Power BI dashboard.

3. Project Objectives

- Build a clean, interview-ready end-to-end analytics project covering data cleaning, SQL analysis, data modeling, and BI reporting
- Identify the top 3 departments with the highest attrition rate and underlying drivers
- Analyze the relationship between compensation, job satisfaction, and employee turnover
- Track headcount by department, gender, education level, and job role
- Provide HR leadership with a 3-page interactive Power BI dashboard with slicers and drill-throughs
- Document all SQL queries and Python scripts for reproducibility and GitHub publication

4. Scope

4.1 In Scope

- Data ingestion and cleaning of the IBM HR Analytics dataset using Python (Pandas)
- SQL analysis layer in PostgreSQL — minimum 10 business queries with documented outputs
- Data modeling in Power BI — star schema with fact and dimension tables
- DAX measures for all KPIs defined in Section 6
- 3-page Power BI dashboard: Overview, Attrition Deep Dive, Salary Analysis
- GitHub repository with README, Python scripts, SQL file, and dashboard screenshots
- Business insights document summarizing 5-7 key findings

4.2 Out of Scope

- Predictive ML model for attrition (deferred to future phase)
- Real-time data integration or live API connections
- Employee-level PII or personally identifiable information
- Mobile-optimized Power BI report layout

5. Stakeholders

Stakeholder	Role	Expectation
HR Director	Primary Business Owner	Dashboard showing attrition trends and salary benchmarks by dept
HR Business Partners	End Users	Ability to filter by department, job role, and gender to identify at-risk groups
Finance Team	Secondary Stakeholder	Headcount and salary cost visibility by department
Yash Awasthi (Analyst)	Project Owner & Developer	End-to-end project showcasing Python, SQL, Power BI, and data storytelling skills

6. Key Performance Indicators (KPIs)

The following KPIs will be tracked and displayed on the dashboard:

KPI Name	Definition	Target / Benchmark
Overall Attrition Rate	Employees who left / Total employees x 100	Reduce to < 12%
Attrition by Department	Attrition count grouped by department	Identify top 3 at-risk depts

KPI Name	Definition	Target / Benchmark
Total Headcount	Count of active employees at any point	Track by dept & role
Average Monthly Income	Mean salary across all employees or filtered group	Benchmark by job level
Attrition by Age Group	Attrition segmented by age bracket (18-25, 26-35, etc.)	Identify highest-risk age group
Job Satisfaction Score	Avg satisfaction rating (1-4 scale) by department	Flag depts scoring < 2.5
Overtime Impact	% of attrited employees who worked overtime	Benchmark vs non-attrited
Work-Life Balance Score	Avg WLB score for attrited vs retained employees	Compare attrited vs retained
Years at Company	Average tenure of attrited employees	Flag < 2 year patterns
Training Times Last Year	Avg training count for attrited vs retained	Identify training gaps

7. Dataset Description

7.1 Source

- Name: IBM HR Analytics Employee Attrition & Performance Dataset
- Source: Kaggle (publicly available, created by IBM data scientists)
- URL: <https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>
- License: Database Contents License (DbCL) v1.0 — free for personal and academic use

7.2 Dataset Overview

Property	Value
Total Rows	1,470 employee records
Total Columns	35 attributes
Target Variable	Attrition (Yes / No)
File Format	CSV (.csv)
Missing Values	None (clean dataset)
Time Period	Cross-sectional snapshot (no date column)

7.3 Key Columns

Column Name	Data Type	Description
Attrition	Categorical	Target variable — Yes/No (employee left the company)
Age	Integer	Employee age in years
Department	Categorical	HR, R&D, or Sales
JobRole	Categorical	9 distinct job roles
MonthlyIncome	Integer	Monthly salary in USD
JobSatisfaction	Integer (1-4)	Employee satisfaction score
OverTime	Categorical	Yes/No — whether employee works overtime
WorkLifeBalance	Integer (1-4)	Self-reported work-life balance score
YearsAtCompany	Integer	Total tenure in years
EnvironmentSatisfaction	Integer (1-4)	Satisfaction with work environment
TrainingTimesLastYear	Integer	Number of training sessions attended
Gender	Categorical	Male / Female
EducationField	Categorical	Field of employee's education
MaritalStatus	Categorical	Single / Married / Divorced
JobLevel	Integer (1-5)	Seniority level of the employee

8. Functional Requirements

8.1 Data Cleaning & Preparation (Python)

- Load raw CSV and perform data type validation and correction
- Encode Attrition column as binary (Yes = 1, No = 0) for analysis
- Create derived columns: AgeGroup (18-25, 26-35, 36-45, 46+), TenureBracket (0-2 yrs, 3-5 yrs, 6-10 yrs, 10+ yrs), SalaryBand (Low / Mid / High)
- Export cleaned dataset as CSV for SQL import and Power BI connection
- Save EDA visualizations (histograms, boxplots, correlation heatmap) as PNG files

8.2 SQL Analysis Layer (PostgreSQL)

- Create a normalized database schema with a main employee table
- Write and document minimum 10 business queries covering:
 - Overall attrition rate and headcount by department

- Average salary by job role and level
- Attrition rate by overtime status
- Top 5 job roles by turnover
- Satisfaction scores for attrited vs retained employees
- Tenure distribution of employees who left
- Gender-based attrition comparison
- Department-wise monthly income vs attrition correlation
- Training frequency for attrited vs retained employees
- Attrition rate by age group using CASE WHEN

8.3 Power BI Dashboard

- Connect Power BI to cleaned CSV or PostgreSQL database
- Build star schema data model with fact and dimension tables
- Create all DAX measures defined in Section 6
- Design 3-page interactive dashboard:
 - Page 1 — Overview: Total headcount, attrition %, KPI cards, dept breakdown chart
 - Page 2 — Attrition Deep Dive: Slicers by dept/age/gender, bar charts, matrix visual
 - Page 3 — Salary Analysis: Income by job level/role, overtime impact, satisfaction scores
- Apply custom theme JSON for consistent branding
- Add page navigator for multi-page navigation
- Implement at least one drill-through from overview to department detail

9. Non-Functional Requirements

- All Python scripts must run without errors in Python 3.10+
- SQL queries must run on PostgreSQL 14+ and be annotated with comments
- Power BI dashboard must load within 5 seconds on a standard laptop
- Dashboard must remain readable at 80% zoom on a 1080p monitor
- GitHub README must include: business problem, tools used, KPIs, key insights, and screenshots
- Project must be fully reproducible — all raw data, scripts, and config files committed to GitHub

10. Deliverables & Timeline

#	Deliverable	Tool	Target
1	Business Requirement Document (this doc)	Word / PDF	Day 1
2	Data Cleaning Script (ibm_hr_clean.py)	Python / Pandas	Day 2-3
3	EDA Report with visualizations (PNG outputs)	Matplotlib / Seaborn	Day 3-4
4	SQL Analysis File (hr_analysis.sql) — 10+ queries	PostgreSQL	Day 5-7
5	Power BI Data Model (star schema)	Power BI Desktop	Day 8-9
6	DAX Measures file (documented)	Power BI / DAX	Day 9-10
7	3-Page Power BI Dashboard (.pbix)	Power BI Desktop	Day 11-13
8	Business Insights Document (5-7 findings)	Markdown / Word	Day 13-14
9	GitHub Repository with full README	GitHub	Day 14

11. Assumptions & Constraints

11.1 Assumptions

- The IBM HR dataset is treated as real organizational data for the purpose of this project
- All salary figures are in USD and represent monthly income
- The dataset represents a single point-in-time snapshot — no longitudinal or time-series analysis is possible from the base data
- Attrition = Yes means the employee has voluntarily left or been terminated

11.2 Constraints

- Dataset has no date column — time intelligence DAX measures will be limited or simulated
- Dataset contains only 1,470 rows — statistical significance of sub-segment findings should be interpreted cautiously
- No access to real company HR data — all findings are based on synthetic IBM dataset

12. Success Criteria

This project will be considered successful when:

- All 10 KPIs are accurately calculated and visible on the dashboard
- The Power BI dashboard has 3 pages with working slicers, drill-through, and page navigation
- At least 10 SQL queries are documented and produce correct outputs
- Python cleaning script runs end-to-end without errors
- GitHub repository is public, complete, and includes a detailed README with screenshots
- A minimum of 5 business insights are written in plain English referencing specific numbers from the data
- The project can be explained clearly in a 5-minute interview walkthrough

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